

TrueYield recovery report — Clark County-NV - Hollywood Rec Center (system 1419)

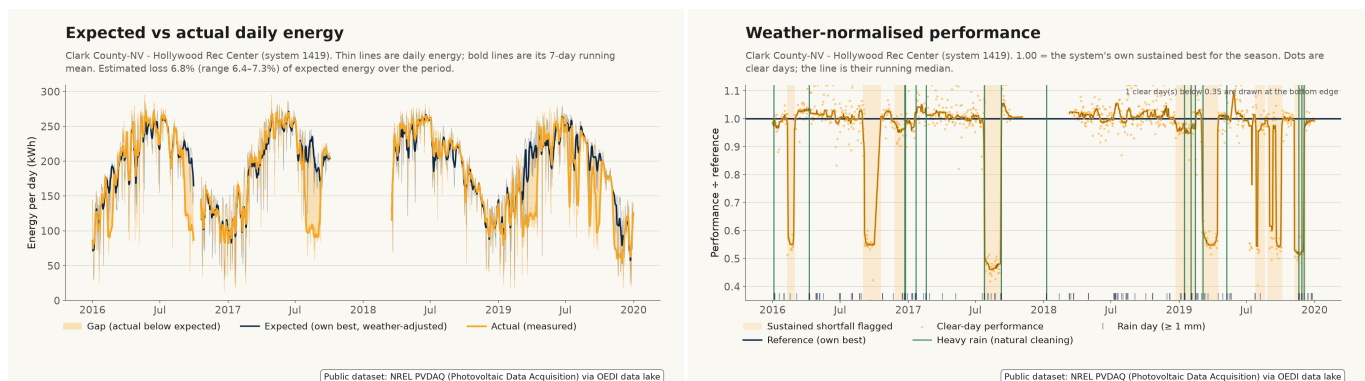
Data label: Public dataset: NREL PVDAQ (Photovoltaic Data Acquisition) via OEDI data lake | Period 2016-01-01 to 2019-12-31 | 40 kWp, tilt 10°, azimuth 210° | generated 2026-10-04

1. What is measured (facts)

- Generation: 1267 valid days of 1461 (86.7% complete); 186 days with no data, 3 partial days and 5 days with impossible values were excluded, not filled in.
- Loader inferred: pvdaq_system_1419.csv: system 1419; time column 'measured_on'; cumulative_counter from ['ac_kwhr_1_kwhac_4798', 'ac_kwhr_2_kwhac_4800'] in kWh; 15-minute steps. 2 counters (one per inverter) are summed. sampling interval varies: 15 min on 958 days, 5 min on 453 days.
- Weather: NASA POWER satellite irradiance (all-sky and clear-sky), 2 m temperature and rainfall for the site coordinates. No on-site sensor is used.

2. What is estimated, and how sure

- Estimated loss: 6.8% of expected energy (range 6.4–7.3%), 16,616 kWh (range 15,525–17,838 kWh) over 1267 valid days. Of this, 16,616 kWh (range 15,525–17,838) is sustained shortfall in 10 flagged episode(s), including 15,552 kWh (range 14,101–16,173) on days more than 25% below the reference, and 0 kWh (range 0–0) is from days with near-zero output. Ranges are 95% bootstrap intervals covering day-to-day noise and the uncertainty of the reference. Shortfalls below the trigger are counted as zero, so this is a conservative figure.
- Detection limit: losses below about 5.1% sustained over 14 days, or about 4.0% sustained over 30 days, cannot be distinguished from noise (from clear-day scatter of 3.9% per day; 1% false-alarm rate per window, 80% power, starting from a system at its reference). Checked with injected-loss trials on real data from system 1420: reliable detection (at least 80% of trials) was reached at about 4% over 14 days, and at about 4% over 30 days. Smaller losses are counted as zero.
- Method: Hourly mode. NASA POWER hourly global irradiance is split into beam and diffuse (Erbs), transposed to the array plane (Hay-Davies, pvlib) and converted to DC energy with a PVWatts model (temperature coefficient -0.4%/°C, NOCT 45 °C). Hourly values are summed to days. Performance index = measured ÷ modelled energy. The reference is the system's own 90th-percentile sustained (7-clear-day median) performance per calendar month, pooled over all years and reduced by the upward bias expected from noise. Clear day = all-sky ÷ clear-sky irradiation ≥ 0.85. Assumptions: constant inverter and wiring losses are absorbed by the reference; timestamps are local standard time; the cleanest state seen in each month is representative of a clean array.



3. Likely causes: detected facts vs hypotheses

Cause	Detected facts	Hypothesis (not proven)
outage or fault indicated	No day with good irradiance had near-zero output. 186 day(s) have no usable generation data (nothing logged, or a reading missing from at least one inverter); whether the system produced on those days is unknown and they are not counted as loss. Sustained output was more than 25% below the reference on 188 day(s) (median level 55% of reference): 2016-02-12 to 2016-02-27 (16 days); 2016-09-05 to 2016-10-08 (34 days); 2017-07-25 to 2017-09-07 (45 days); 2019-03-05 to 2019-04-14 (41 days); 2019-07-28 to 2019-07-31 (4 days); 2019-09-02 to 2019-09-10 (9 days); 2019-09-20 to 2019-10-03 (14 days); 2019-11-08 to 2019-12-02 (25 days). This accounts for 15,552 kWh of the sustained shortfall.	The deep sustained drop is consistent with part of the system being off (one inverter, string or meter channel), which is too large and too abrupt for soiling.
soiling not indicated	Between cleaning events, performance declined in 12 of 12 usable intervals, at a median of 0.349% per day in the declining ones. 0 of 9 testable natural rain event(s) were followed by a rise in normalised performance that is distinguishable from zero. rdtools estimates a soiling loss of 5.9% (range 3.3–9.1%, rdtools 3.2.1 stochastic rate and recovery (SRR)). Shuffle check: with the same clear-day values in random time order, which contain no soiling pattern, it reports 2.7–5.1%.	No soiling distinguishable from scatter. The estimate of 5.9% (range 3.3–9.1%, rdtools 3.2.1 stochastic rate and recovery (SRR)) is not clearly above what the same method reports on shuffled data (up to 5.1%), so most or all of it can be produced by day-to-day scatter alone.

ageing or degradation not indicated	Year-on-year change of clear-day performance over 4.0 years: -0.44% per year (range -1.00 to +0.13% per year, rdtools 3.2.1 year-on-year).	No ageing trend distinguishable from zero: the range includes zero or is positive.
shading cannot be assessed	Clock check: on clear days the measured daily profile is centred +51 minutes from the modelled one (+57 min in May-August, +8 min in November-February). The timestamps do not line up with solar time well enough for an hour-by-hour test (the summer-winter difference suggests the logger follows daylight-saving time).	—

4. Recovery verification

- Events tested: natural rain event, not a maintenance intervention. 9 of 17 event(s) had enough clear days on both sides (14 days before, 14 after).
- Best-observed event, 2019-01-15 (natural rain event, not a maintenance intervention): -1.2% (range -5.0 to +1.6%), -1.5 kWh/day (range -5.8 to +1.8), from 7 clear days before and 9 after. It was chosen for having the most clear days, not for its size. The range includes zero: no recovery can be claimed for this event.
- Across all testable events: 0 clear rise(s), 0 clear fall(s), 9 not distinguishable from zero. Full list in summary.json.

5. Value and verdict

- [not available: tariff_inr_per_kwh not provided in data/systems.csv]
- Emissions: [not available: emission_factor_kg_per_kwh is empty in config.yaml; no factor is assumed]

6. Limits of this report

- Causes are hypotheses; none is proven without a site inspection.
- Loss is measured against this system's own best month by month; a loss present from the start, or one that recurs in the same month every year, is partly or wholly invisible.
- Irradiance is a satellite area average; losses below the detection limit are not counted.
- 188 day(s) in deep-drop episodes (part of the system off) are excluded from the recovery, soiling, ageing and shading tests, so those tests describe the system while it was otherwise working.

7. Method validation with injected loss — not a field result

- On a clean real stretch of system 1420 (2018-04-13 to 2018-12-08), a 10% soiling ramp and a 3-day outage were injected (1,221 kWh). The unchanged pipeline recovered 1,041 kWh (85% of the injected loss; estimate 1,041 kWh, range 762–1,610); the known truth is inside the stated range. Outage days found: 3 of 3. Detection-limit check: over 14 days: 2% step found in 27%, 4% step found in 80%, 6% step found in 93%, 8% step found in 97%, 10% step found in 100%, 12% step found in 100%; over 30 days: 2% step found in 47%, 4% step found in 90%, 6% step found in 100%, 8% step found in 100%, 10% step found in 100%, 12% step found in 100% of 30 trials each. These are test results on altered data, not field results.